Arnav Poswal

Bachelors in Computer Science and Engineering
SRM University, Kattankulatham, Chennai

+91-80597 42431
poswalarnav37@gmail.com
LinkedIn | GitHub

Education

•B.Tech – Computer Science and Engineering *2021–2025*
SRM University, Kattankulatham, Chennai CGPA: 8.33 / 10

Experience

•Software Developer *Mar 2025 – Present*
Birbals Inc.

- Shipped AI features to production: built a multi-level RAG pipeline using LangChain and LlamaIndex, an AI research agent, and the integration of multiple LLMs (Anthropic, OpenAI) on a vehicle-management platform and a pet health platform.
- Led a 2-person team to migrate company-wide CI/CD from Jenkins to GitHub Actions, standardizing Docker-based automated deployments across all projects.
- Built a centralized ELK logging pipeline ingesting 20 GB/day from 18 services, providing every team with one place to monitor errors across environments.
- Designed and implemented a file management service using Next.js, Node.js, and AWS S3, reducing manual document uploads for developers by 70%.
- Collaborated with design and product management teams to align technical solutions with customer needs; contributed to both technical design documentation and end-user documentation.

Projects

• OrderFlow: Event-Driven Order Platform
An event-driven order platform for grocery-style orders, built as independent microservices that coordinate through Kafka.

- Architected three Spring Boot microservices (order, inventory, payment), each with its own PostgreSQL database, coordinating through Kafka with a choreographed saga and compensation events (stock release and refund on failure).
- Ensured consistency with a transactional outbox, idempotent consumers, per-order Kafka keys, and atomic stock updates; used Redis for idempotency keys and caching.
- Validated concurrency correctness with a k6 race test (50 concurrent orders for 1 unit of stock): exactly one order confirmed and zero overselling; added Testcontainers integration tests, 9 acceptance scenarios, and GitHub Actions CI.

Tools: Java 21, Spring Boot, Kafka, PostgreSQL, Redis, Flyway, Testcontainers, Docker, Next.js, ELK, k6.

• FocusFleet: AI-Powered Driver Drowsiness Detection
A real-time driver drowsiness detection system that detects fatigue patterns from live camera streams.

- Designed and deployed a real-time drowsiness detection system processing live camera streams at 30fps — trained a TensorFlow/Keras model on Eye Aspect Ratio (EAR) and yawn pattern features achieving 96% test accuracy in driver fatigue detection.
- Implemented full-stack features including driver auth, session logging, and a Gradio UI with live camera feed and status dashboards; persisted session data in SQLite.
- Wrote unit tests and end-to-end pipeline validation for the detection pipeline.

Tools: Python, TensorFlow/Keras, MediaPipe, OpenCV, Gradio, SQLite.

• Mobile Fraud Detection
A real-time detection system using XLNet Transformer Models achieving 90% accuracy.

- Improved fraud detection performance by 40% over baseline models using contextual embeddings from XLNet for advanced language modeling; integrated IBM Watson Cloud APIs for scalable training and inference on large-scale speech and text datasets.
- Optimized the data preprocessing pipeline, accelerating model convergence by 15% and reducing feature dimensionality by 10% for faster training and improved efficiency.

Tools: Python, PyTorch, NLP, IBM Watson.

Leadership & Activities

•Coordinated the SRM Tech Club's AI Workshop, mentoring over 50 peers on AI, ML, NLP, and data analysis; effectively conveyed technical concepts to diverse audiences.

Technical & Analytical Skills

Languages: JavaScript/TypeScript, Python, Java, SQL, C/C++
Backend: Node.js/Express, REST APIs, Spring Boot, Microservices, Kafka, PostgreSQL, Redis
AI/LLM: RAG, LangChain, LangGraph, LlamaIndex, Anthropic & OpenAI APIs, NLP
Infra & Tools: AWS (S3, EC2), Docker, GitHub Actions, ELK, Grafana, Git
ML & Data: TensorFlow, OpenCV, Pandas, NumPy
Frontend: Next.js
Concepts: Data Structures & Algorithms, System Design, CI/CD